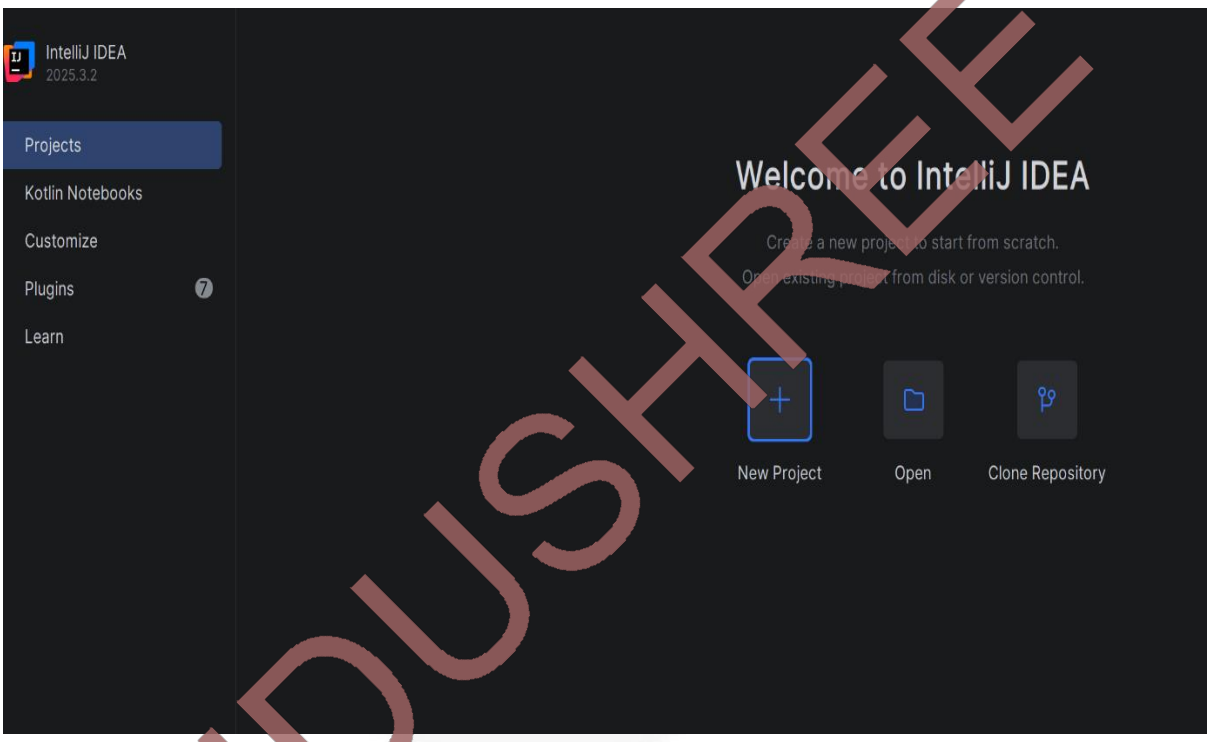
Program 3: Working with Gradle: Setting Up a Gradle Project, Understanding Build Scripts (Groovy and Kotlin DSL), Dependency Management and Task Automation

Step 1: Create a Gradle Project in IntelliJ IDEA

1. Open **IntelliJ IDEA and** Click **New Project**

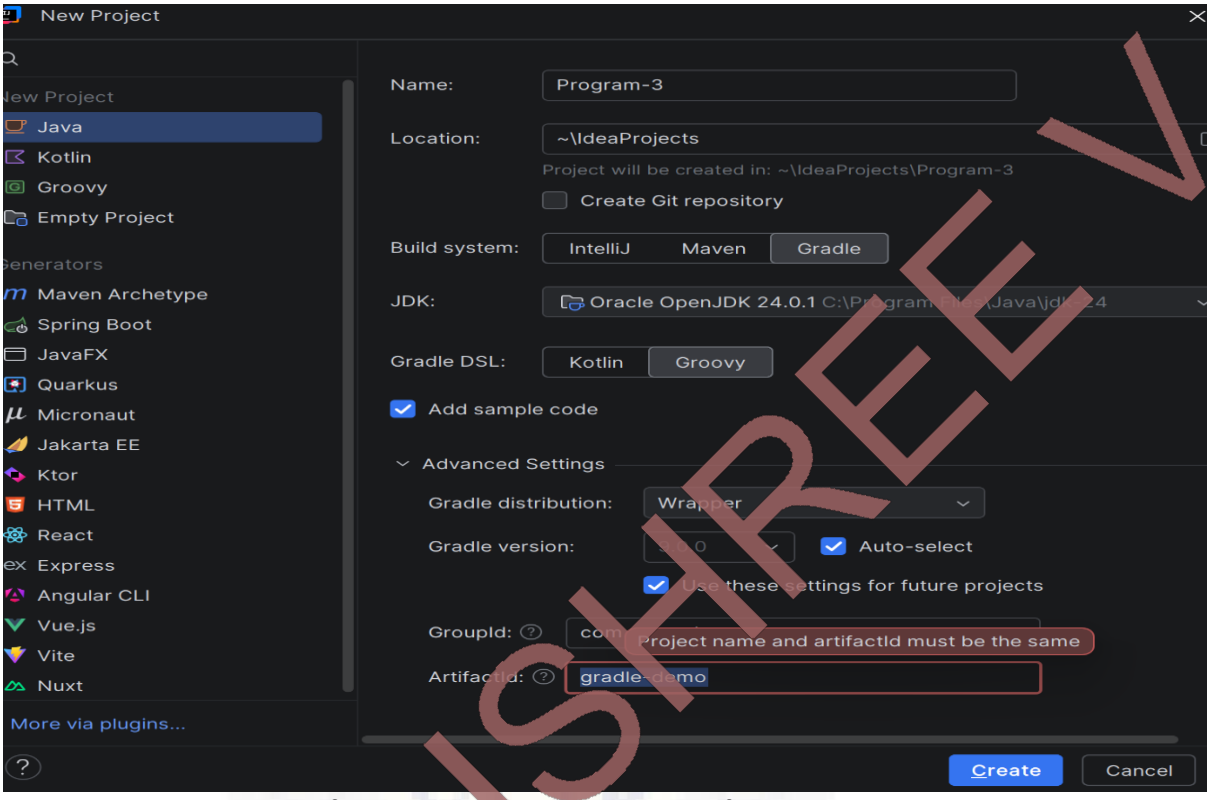


2. Select **Java**
3. **Build Script DSL → Groovy**

   Enter: **Group ID**: com.example

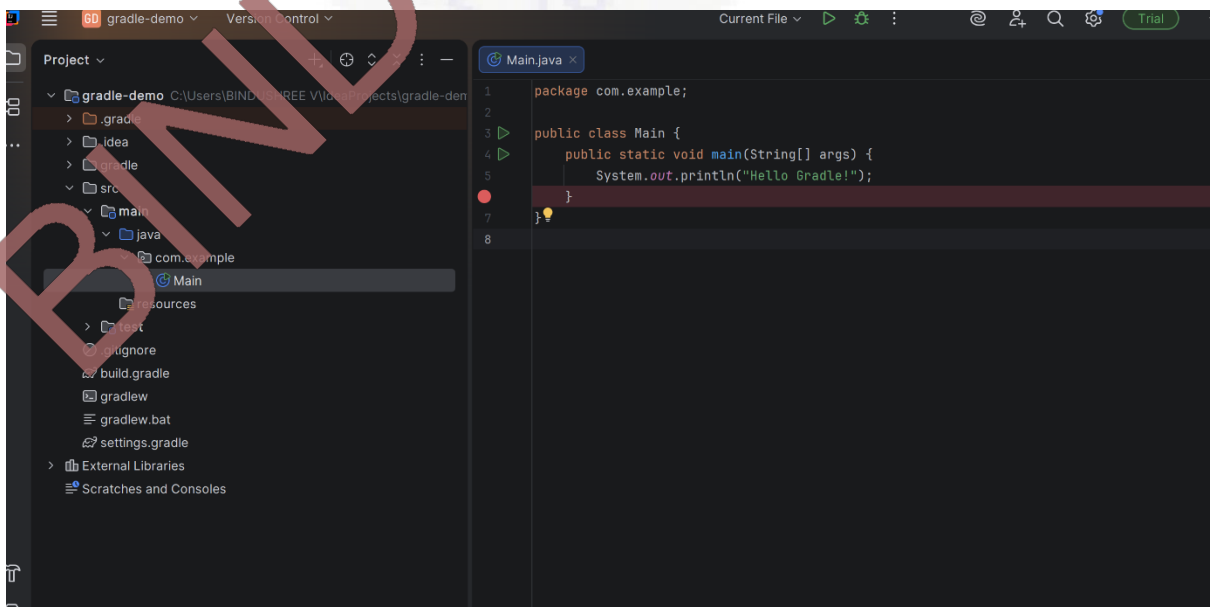
   Artifact ID: gradle-demo

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Note: ArtifactId should be same as that of the Project Name

4. Click **create,** IntelliJ will create the project structure automatically.



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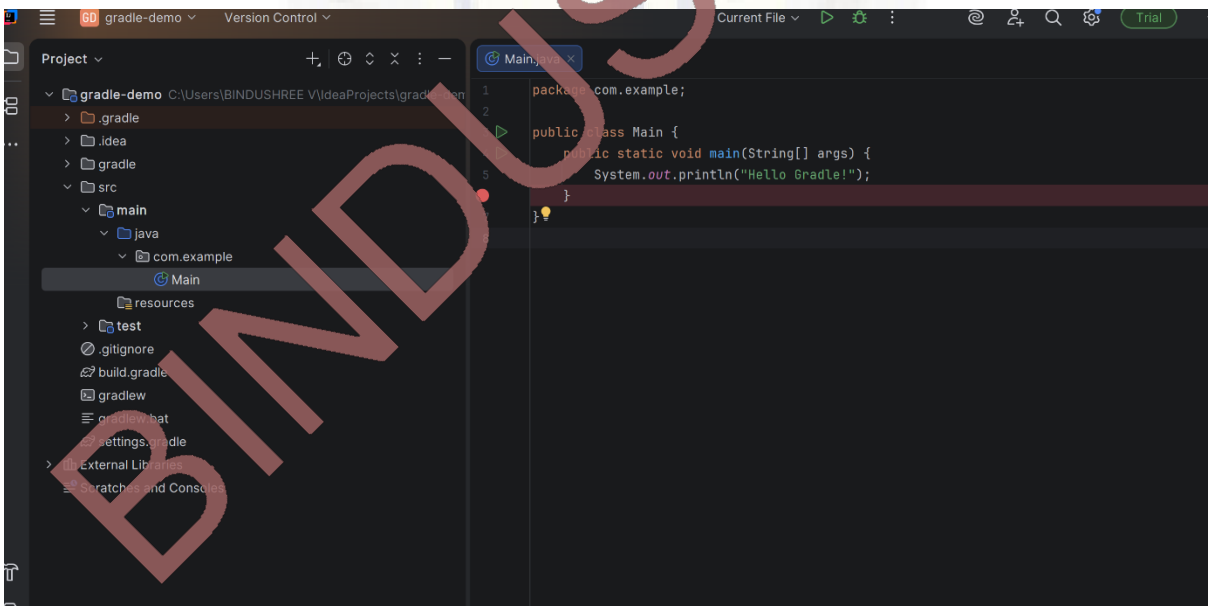
✓ **Gradle Project Structure Created Automatically**

5. Create Main Java File

Open src/main/java/com/example/Main.java

```
package com.example;


public class Main {
    public static void main(String[] args) {
        System.out.println("Hello Gradle!");
    }
}
```



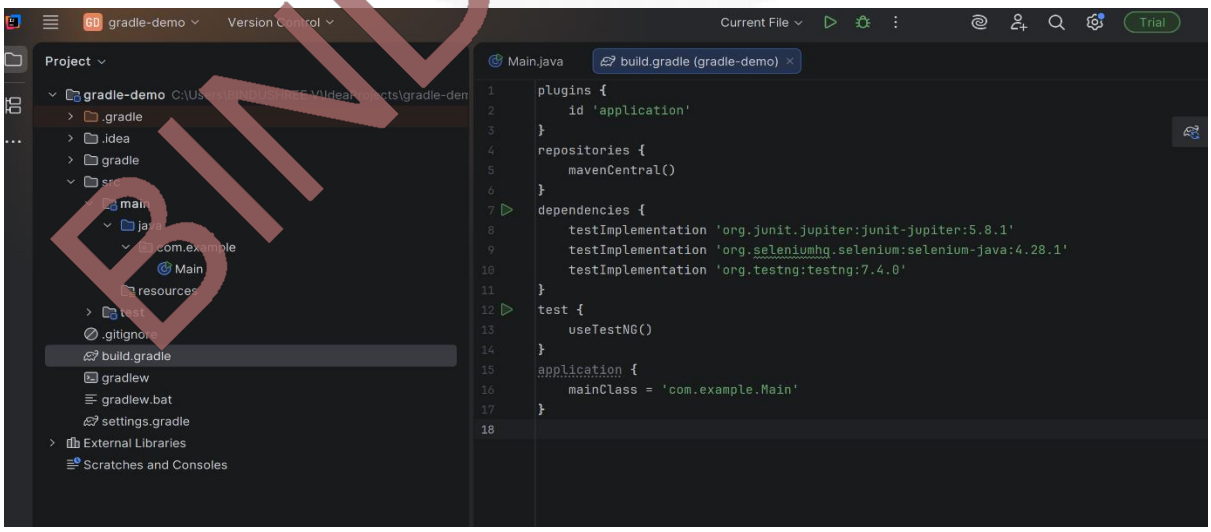
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6. Update build.gradle File

Open **build.gradle** and paste **exactly this**:

plugins {

```
    id 'application'
}
repositories {
    mavenCentral()
}
dependencies {
    testImplementation 'org.junit.jupiter:junit-jupiter:5.8.1'
    testImplementation 'org.seleniumhq.selenium:selenium-java:4.28.1'
    testImplementation 'org.testng:testng:7.4.0'
}
test {
    useTestNG()
}
application {
    mainClass = 'com.example.Main'
}
```



Build gradle updation

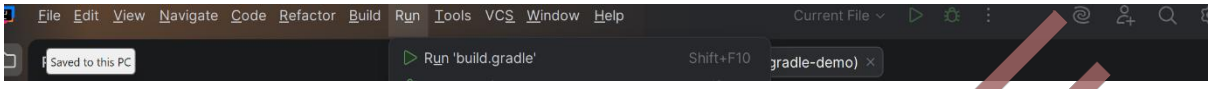
7. Run the Application

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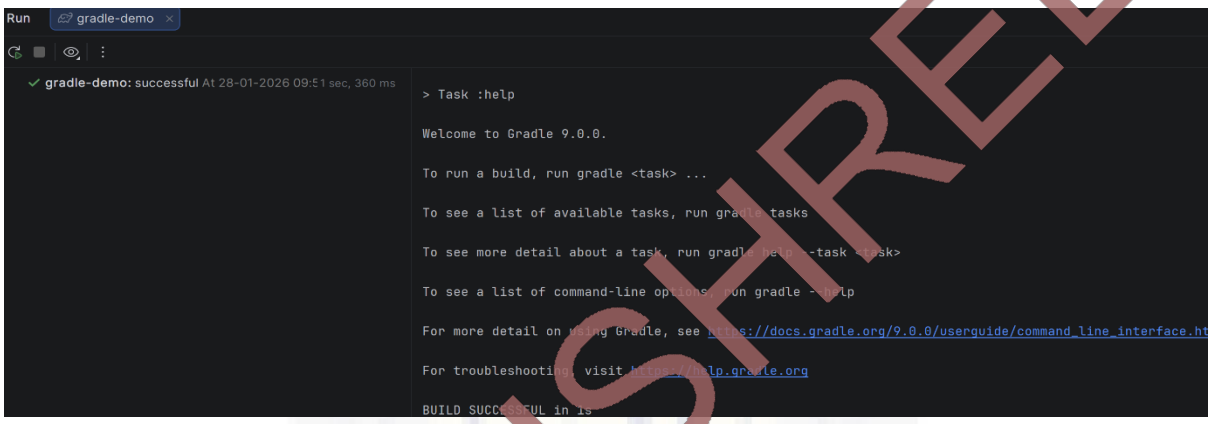
Method 1: Using Terminal

Open terminal in IntelliJ and type:

gradle run



Output:

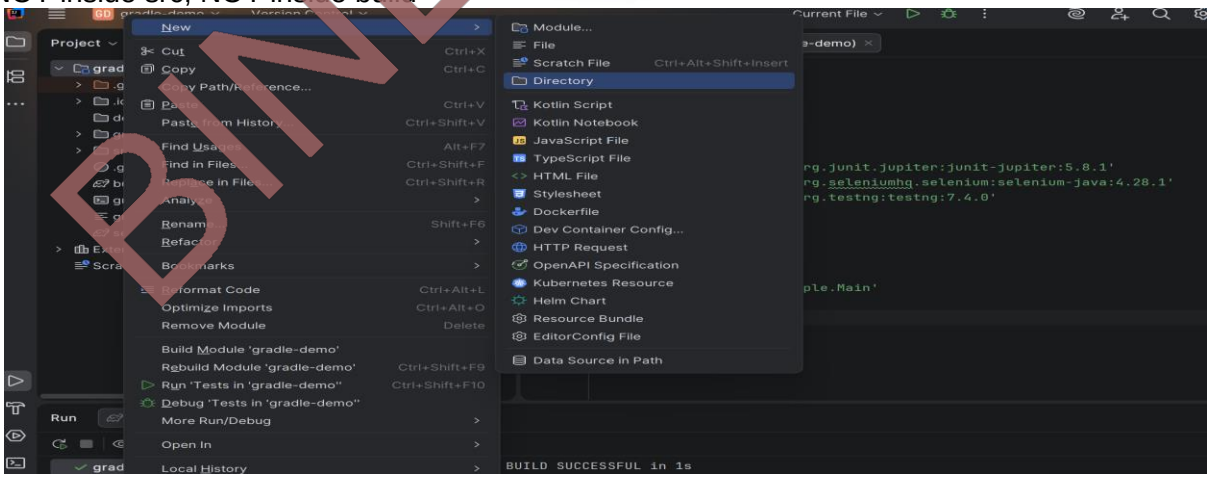


Gradle project is working

8. Create docs Folder (VERY IMPORTANT)

Where to create the docs folder?

The docs folder must be created inside the main Gradle project folder
NOT inside src, NOT inside build



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File 1: index.html

Create this file inside the docs folder

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <title>Tripillar Solutions</title>
   <link rel="stylesheet" href="style.css">
</head>
<body>
   <div class="container">
      <img src="logo.png" alt="Logo" class="logo">
      <h1>Tripillar Solutions</h1>
      <p>Welcome to our official website</p>
   </div>
</body>
</html>
```

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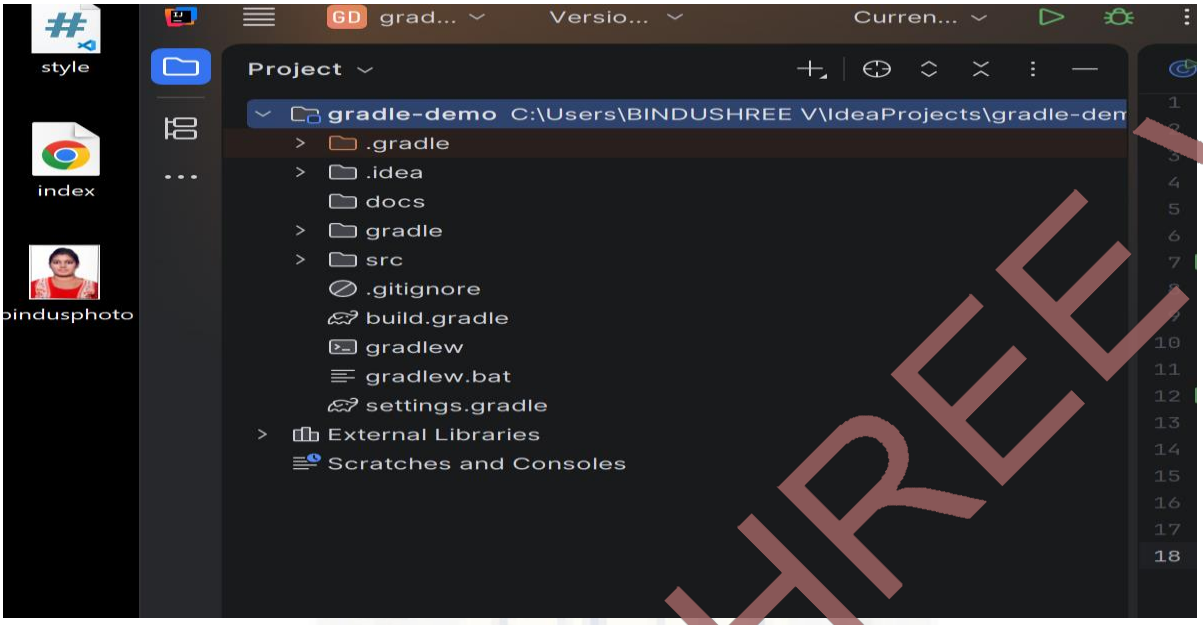
File 2: style.css

Create this file inside the same docs folder

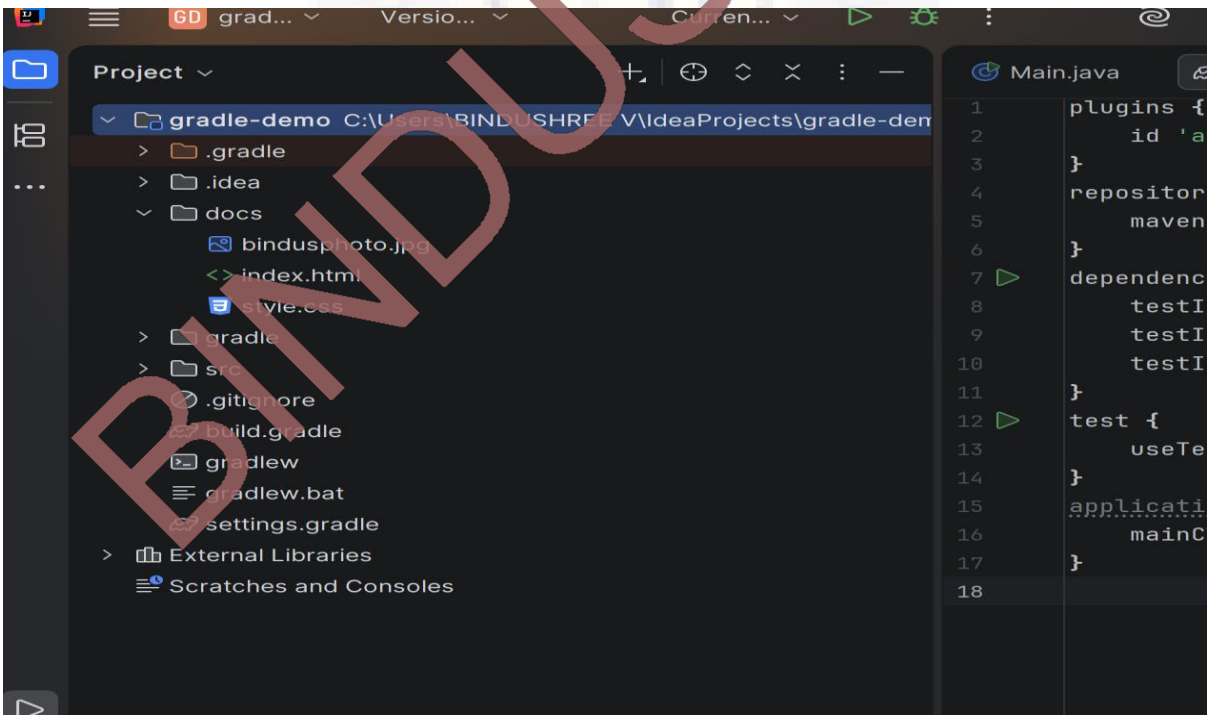
```
body {
    font-family: Arial, sans-serif;
    background-color: #f2f2f2;
}
.container {
    text-align: center;
    margin-top: 100px;
}
.logo {
    width: 150px;
    height: auto;
}
h1 {
    color: #2c3e50;
}
p {
    color: #555;
    font-size: 18px;
}
```

File 3: any .png file

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docs folder has created

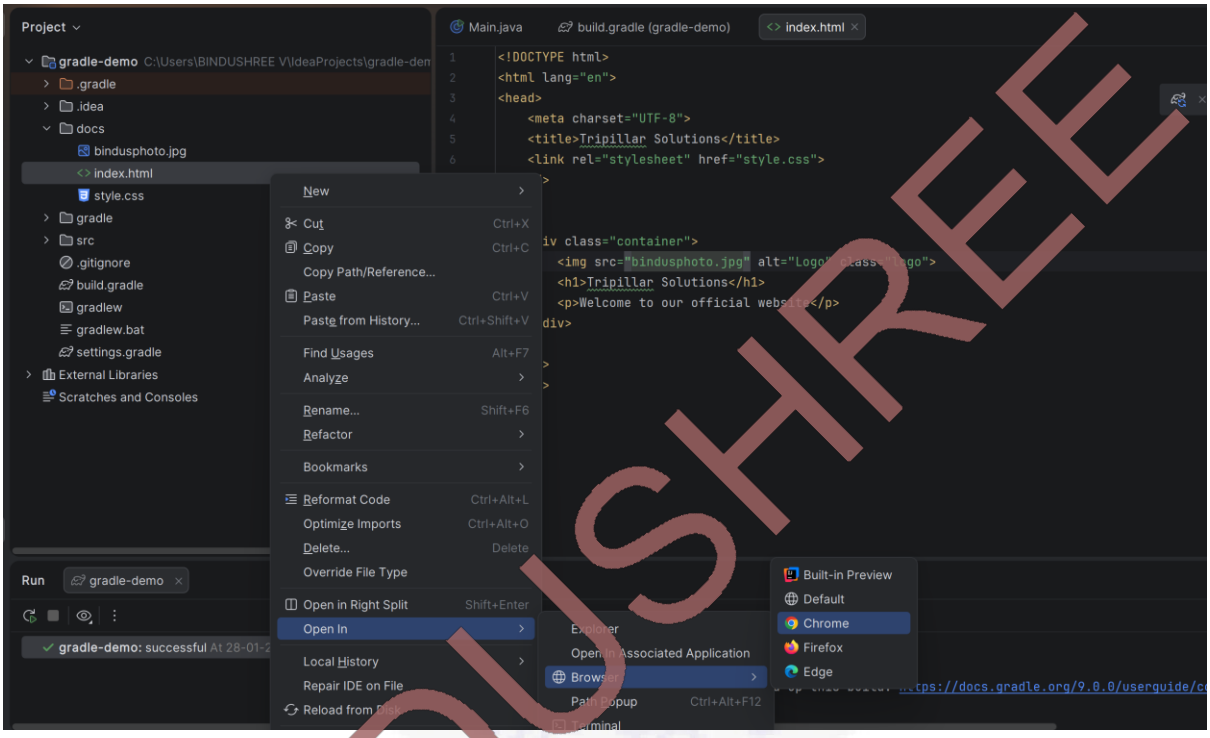


After adding all the above files to docs folder

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9. How to Check if It Works (Before GitHub Pages)

1. Open docs folder in File Explorer
2. Right click **index.html->Open In->Browser->Chrome**



If you can see this:

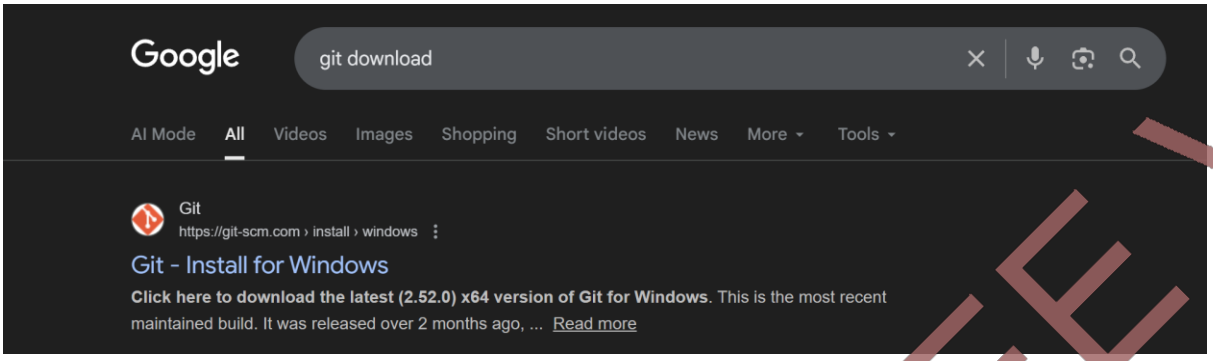
Sample Output on Browser

- Logo image on top
- Heading: **Tripillar Solutions**
- Text: *Welcome to our official website*

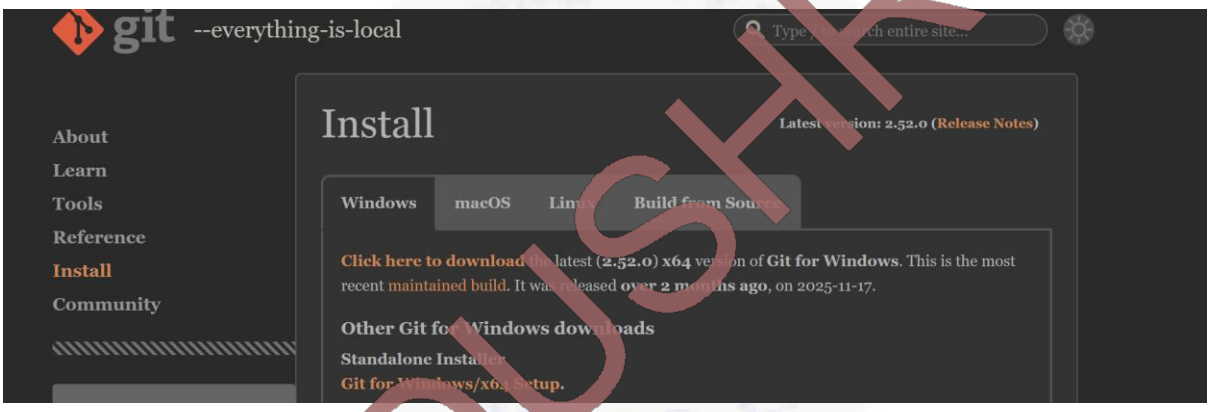
Then its working fine

10. Open Browser & Search for Git

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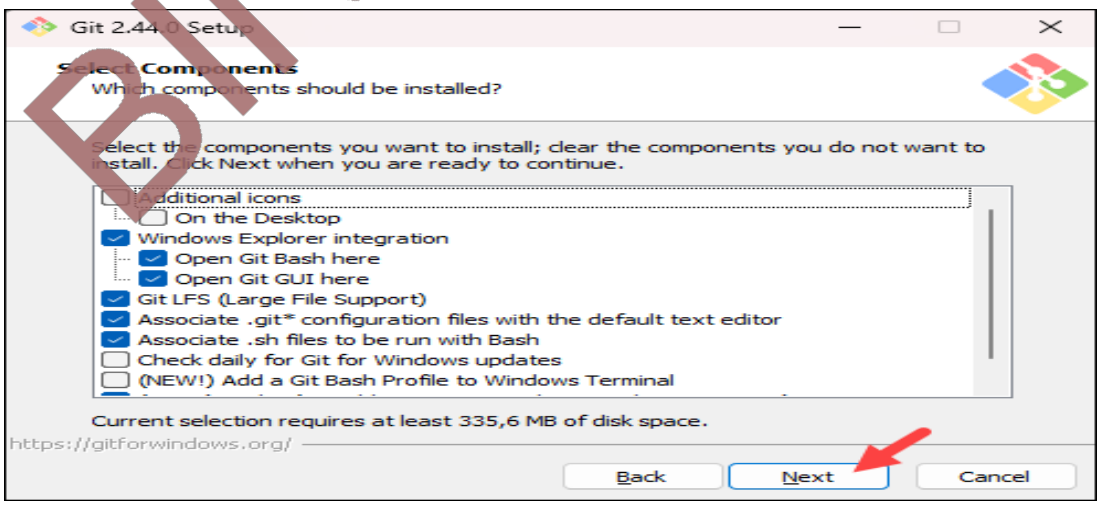


Click the official site: git-scm.com

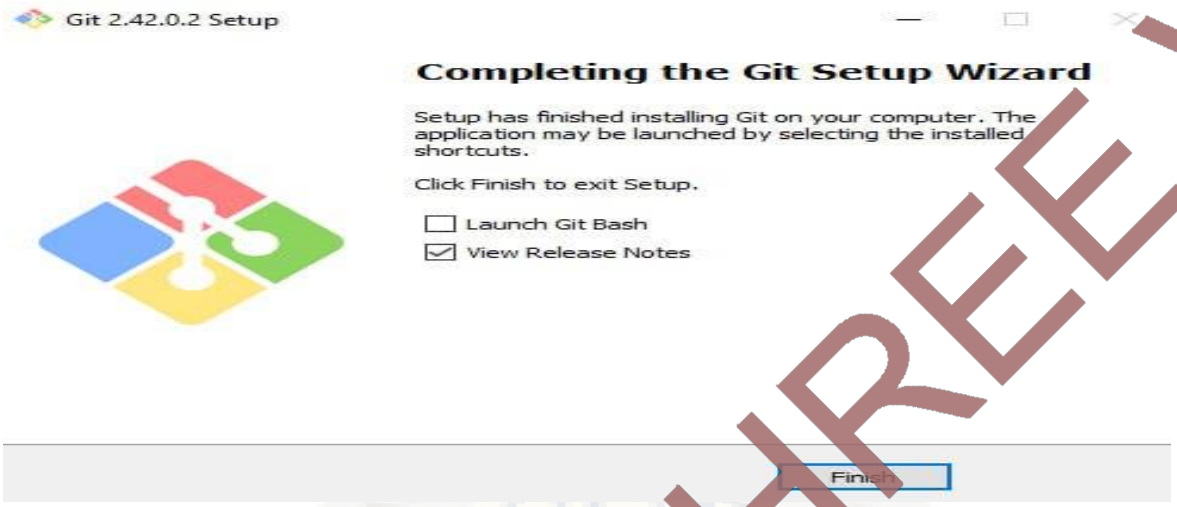


On the website, click: Download for Windows and Download starts automatically

11. Run the installer-> Open Downloads Double->click the downloaded Git installer



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For beginners, **just click "Next"** on all screens (No need to change anything)

Recommended options (Default):

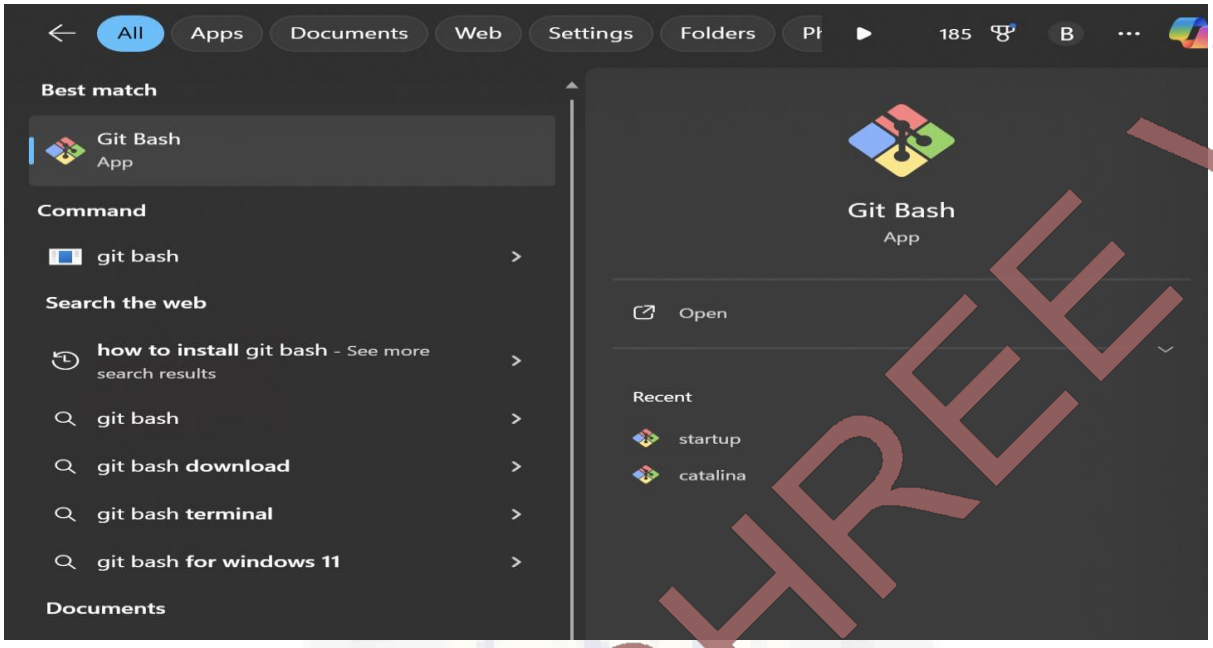
- ✓ Use Git from Git Bash only
- ✓ OpenSSL
- ✓ Checkout Windows style, commit Unix-style
- ✓ Use MinTTY terminal

12. Verify Git Bash Installation

Desktop / Start Menu

Press **Windows Key**
Type: Git Bash

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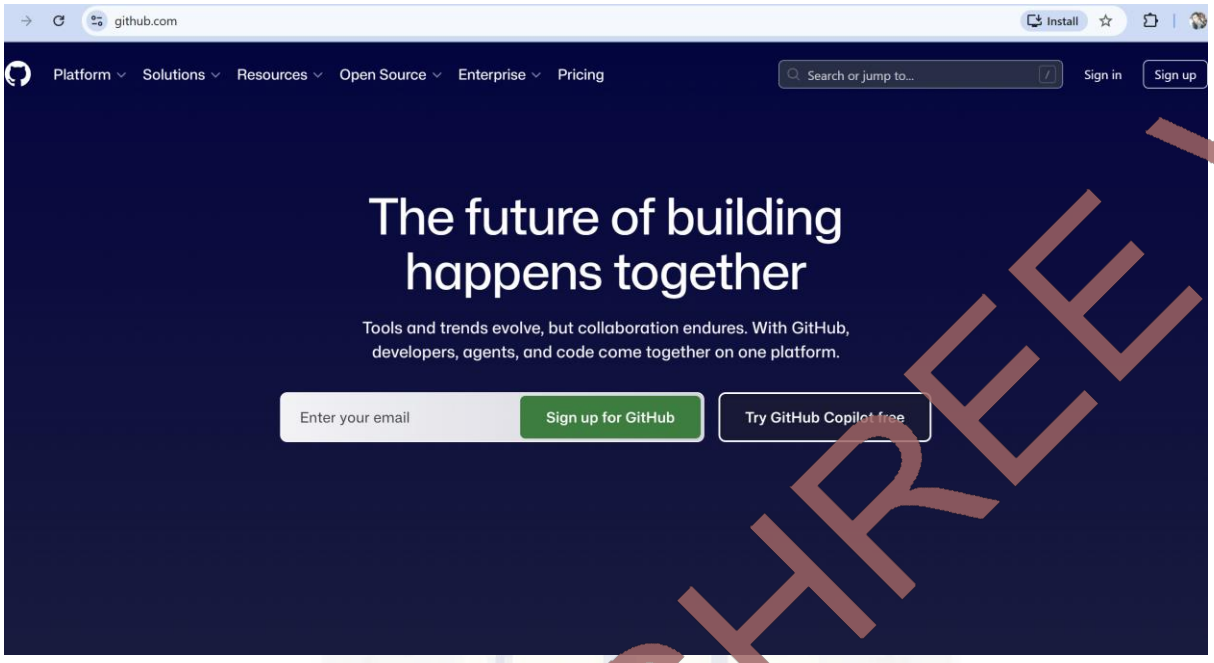


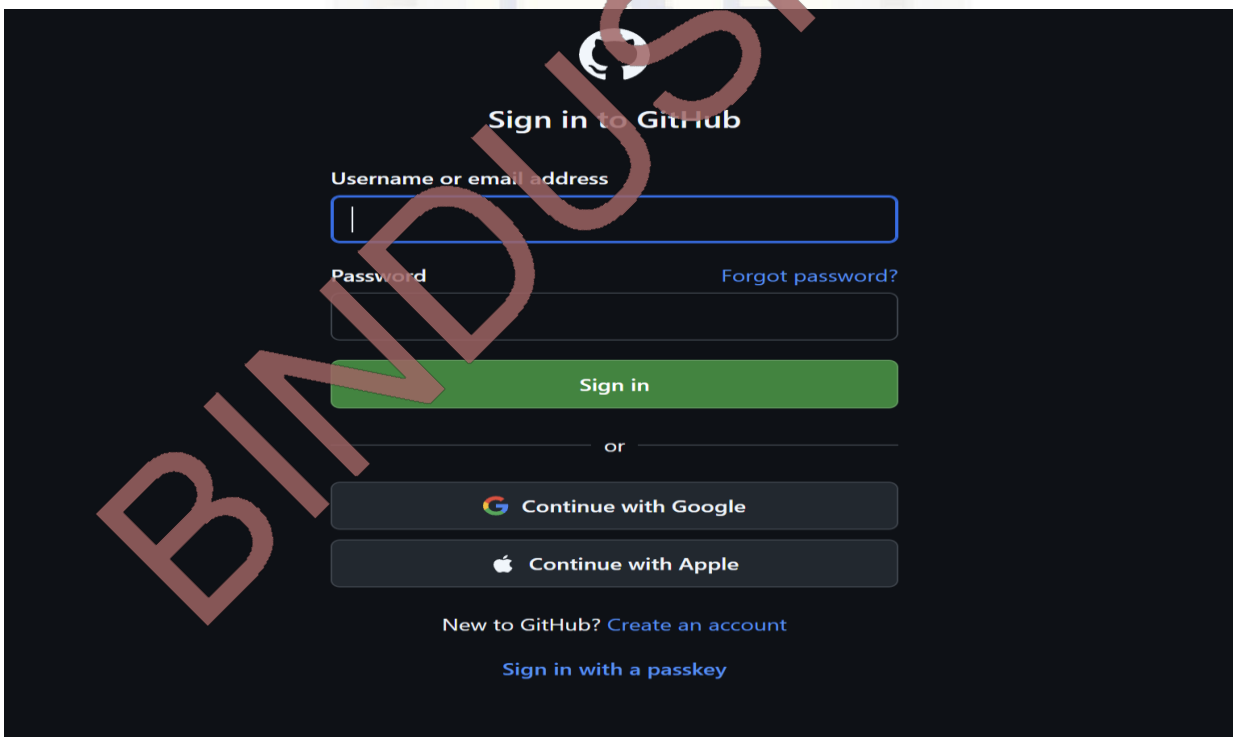


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13. Create a GitHub Account (If not already created)

1. Open browser
2. Go to **github.com**
3. Sign up using email
4. Login to GitHub





Sign in with google

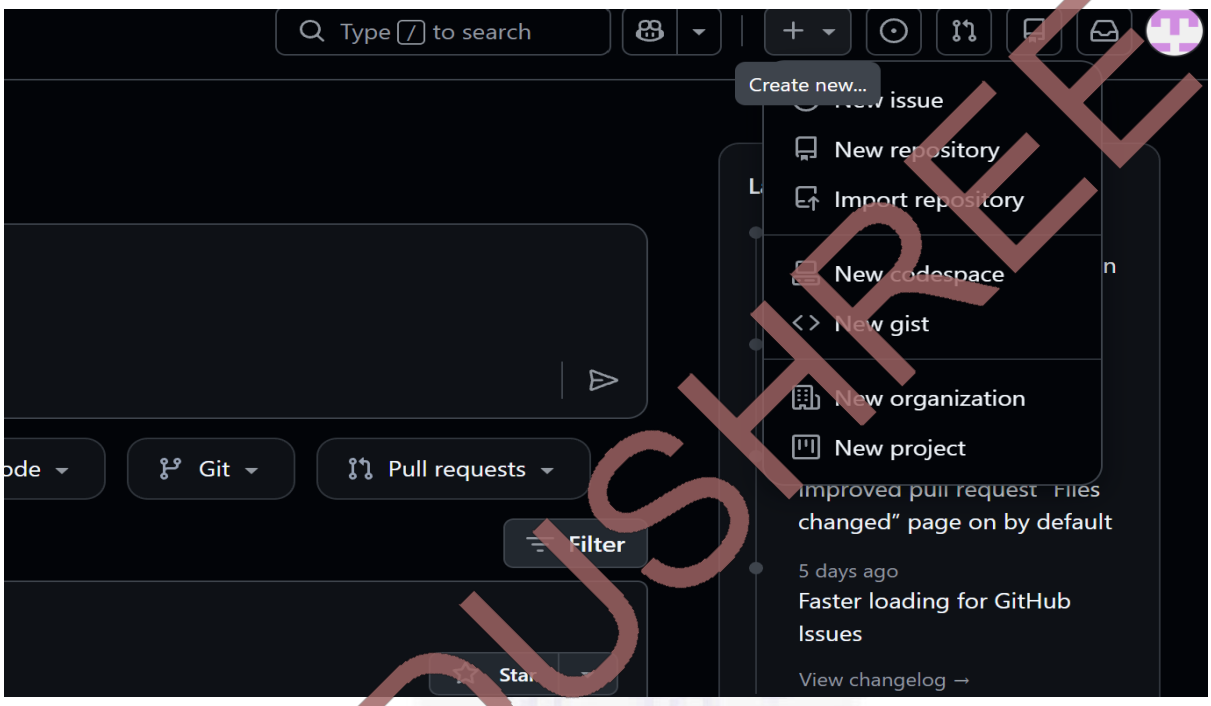
14. Create a New Repository

1. Click **+ (top right corner)**
2. Click **New repository**

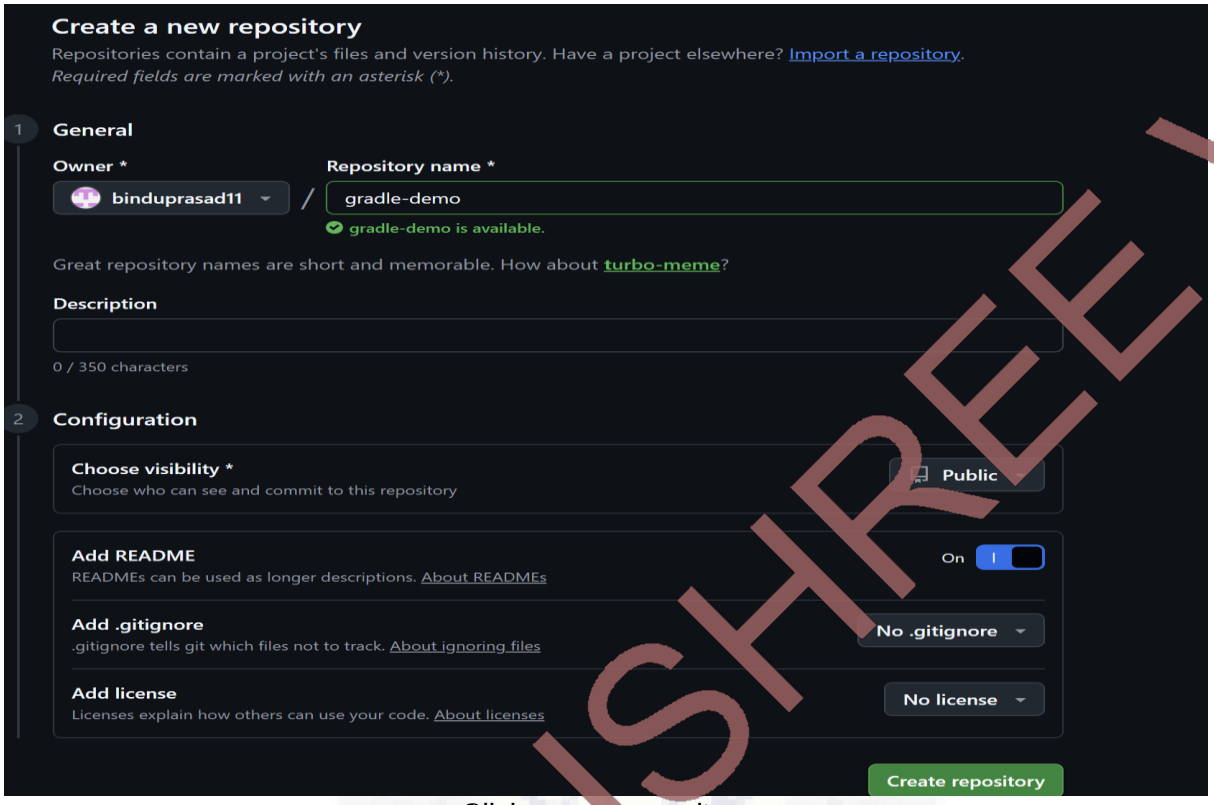
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Fill details:

3. Repository name: gradle-demo
4. Public (important)
5. ✓ Check Add README file
6. Click Create repository



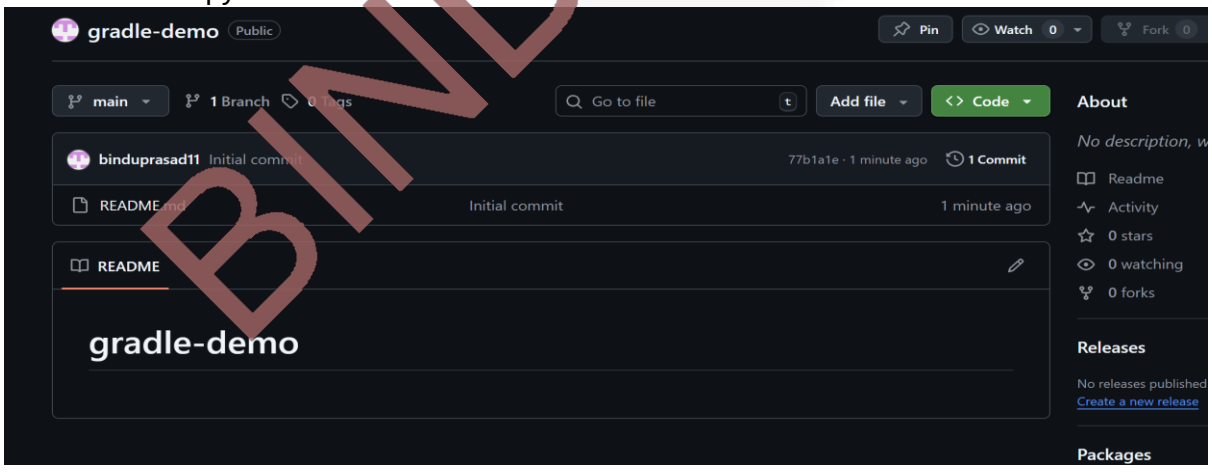
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Click create repository

15. Copy Repository URL

1. Click **Code** button
2. Copy **HTTPS URL**

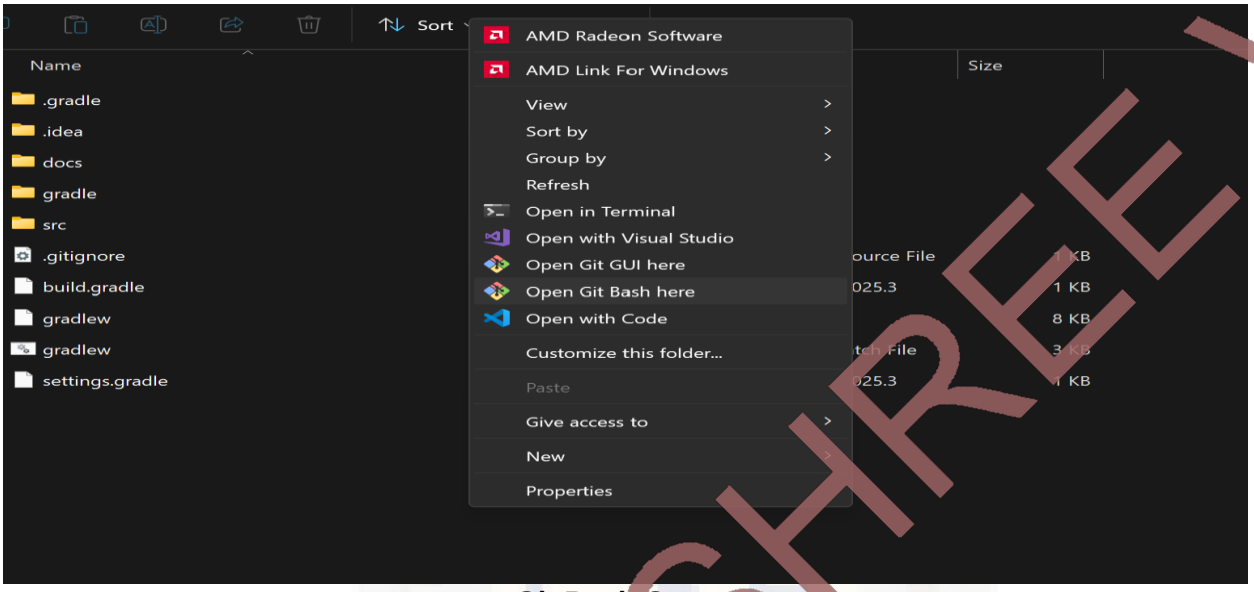


16. Open Project Folder->Open **File Explorer->**Go to your project folder (example: gradle-demo)

Example path in my system: C:\Users\BINDUSHREE V\IdeaProjects\gradle-demo

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Right-Click Inside the Folder Right-click **inside empty space** (not on a file) and Select **Git Bash Here**



Git Bash Opens
You will see a black terminal window.



Type: git init
Check the git status by the command: git status

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```
MINGW64:/c/Users/BINDUSHREE V/IdeaProjects/gradle-demo

BINDUSHREE V@Bindushree MINGW64 ~/IdeaProjects/gradle-demo
$ pwd
/c/Users/BINDUSHREE V/IdeaProjects/gradle-demo

BINDUSHREE V@Bindushree MINGW64 ~/IdeaProjects/gradle-demo
$ git init
Initialized empty Git repository in C:/Users/BINDUSHREE V/IdeaProjects/gradle-de
mo/.git/

BINDUSHREE V@Bindushree MINGW64 ~/IdeaProjects/gradle-demo (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore
        .idea/
        build.gradle
        docs/
        gradle/
        gradlew
        gradlew.bat
        settings.gradle
        src/

nothing added to commit but untracked files present (use "git add" to track)

BINDUSHREE V@Bindushree MINGW64 ~/IdeaProjects/gradle-demo (master)
```

Command: git add .

```
BINDUSHREE V@Bindushree MINGW64 ~/IdeaProjects/gradle-demo (master)
$ git add .
```

Commit the Files: command: git commit -m "Added website files for GitHub Pages"

```
BINDUSHREE V@Bindushree MINGW64 ~/IdeaProjects/gradle-demo (master)
$ git commit -m "Added website files for Github Pages"
[master (root-commit) f2aa15c] Added website files for Github Pages
 14 files changed, 473 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 .idea/.gitignore
 create mode 100644 .idea/gradle.xml
 create mode 100644 .idea/misc.xml
 create mode 100644 build.gradle
 create mode 100644 docs/bindusphoto.jpg
 create mode 100644 docs/index.html
 create mode 100644 docs/style.css
 create mode 100644 gradle/wrapper/gradle-wrapper.jar
 create mode 100644 gradle/wrapper/gradle-wrapper.properties
 create mode 100644 gradlew
 create mode 100644 gradlew.bat
 create mode 100644 settings.gradle
 create mode 100644 src/main/java/com/example/Main.java
```

Connect Local Project to GitHub: url you will get it from step no: 15
Type command: git remote add origin url

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```
BINDUSHREE V@Bindushree MINGW64 ~/IdeaProjects/gradle-demo (master)
$ git remote add origin https://github.com/binduprasad11/gradle-demo.git
```

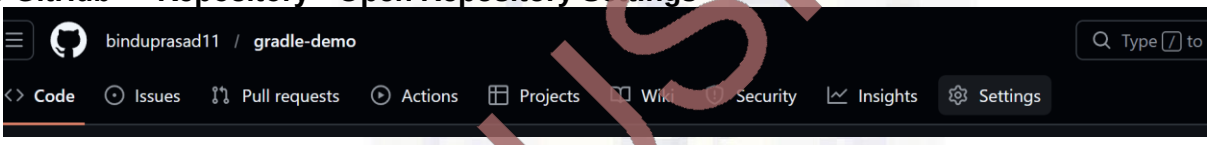
No output = success

Push Files to GitHub

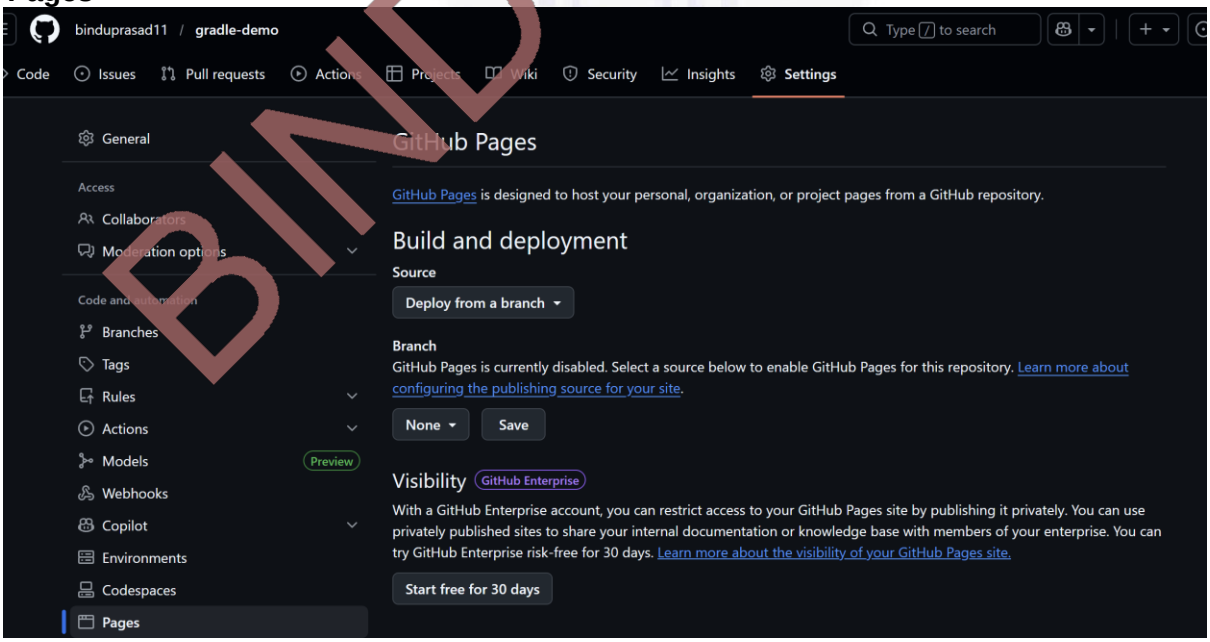
Command: git push -u origin master

```
BINDUSHREE V@Bindushree MINGW64 ~/IdeaProjects/gradle-demo (master)
$ git push -u origin master
Enumerating objects: 25, done.
Counting objects: 100% (25/25), done.
Delta compression using up to 12 threads
Compressing objects: 100% (18/18), done.
Writing objects: 100% (25/25), 155.33 KiB | 5.55 MiB/s, done.
Total 25 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'master' on GitHub by visiting:
remote:      https://github.com/binduprasad11/gradle-demo/pull/new/master
remote:
To https://github.com/binduprasad11/gradle-demo.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
```

Go to GitHub → Repository->Open Repository Settings



Click Pages



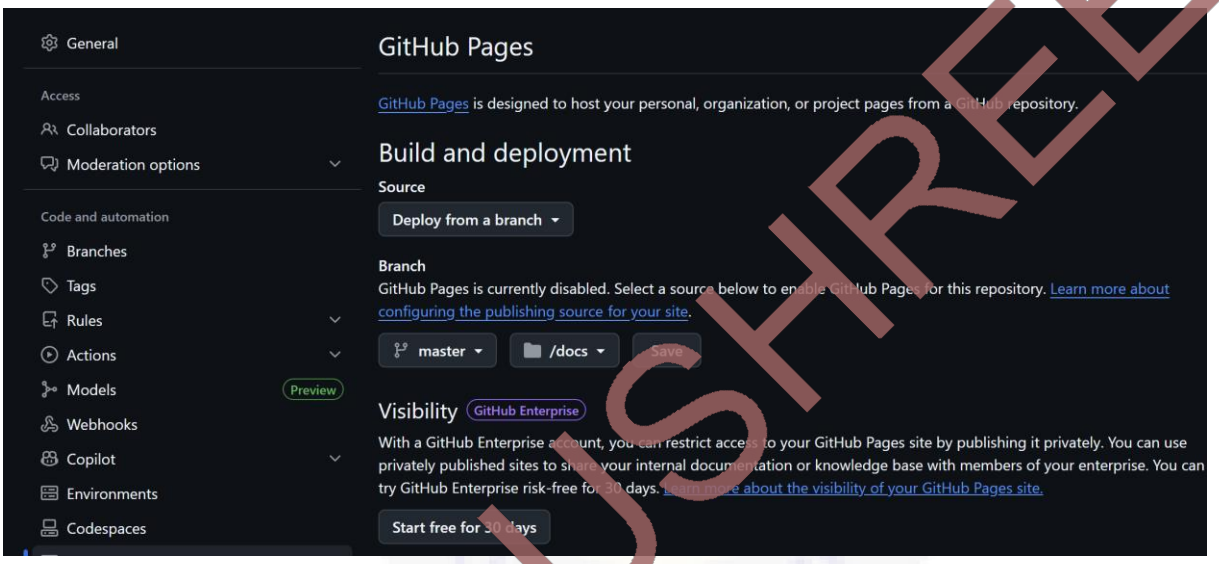
BINDUSHREE V

Select:

- **Branch: master**
- **Folder: /docs**

2. **Click Save**

After 1 minute, website becomes live.



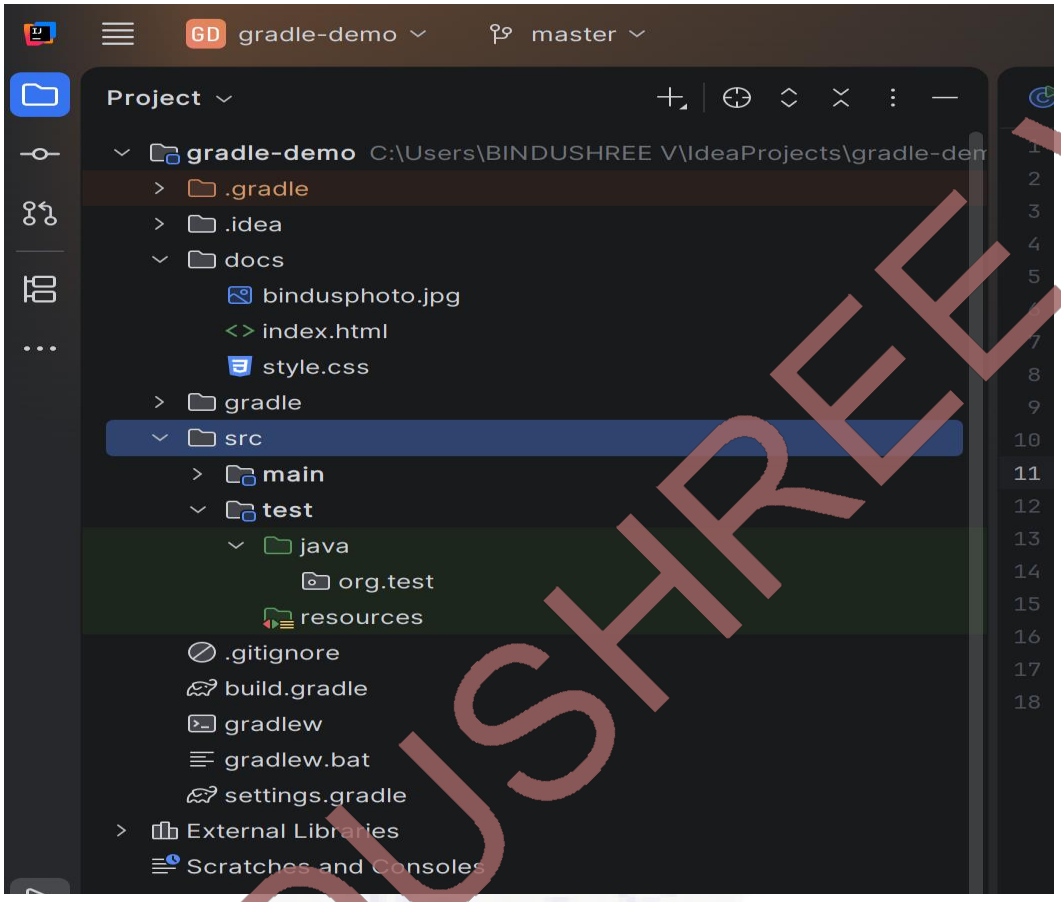
Task 3: Automation Testing of Published Website

Now go to intellij Idea:

- **Right-click src**
- **Click New**
- **Click Directory**
- **In the name field, type: test/java/org/test**

IntelliJ will automatically create all folders.

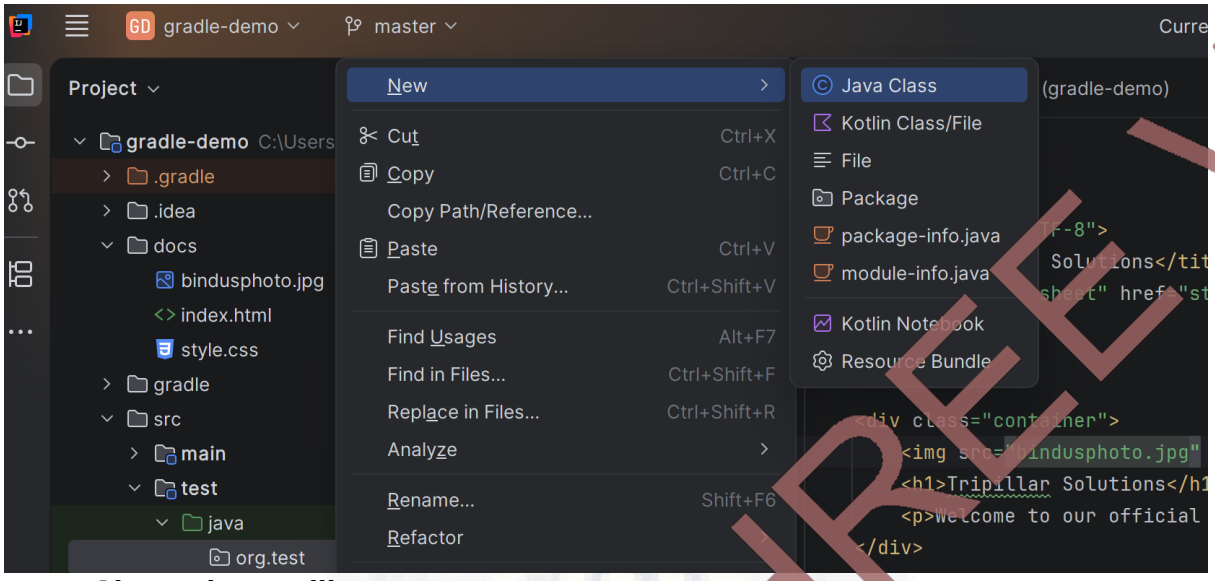
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Create Test Class

Right-click org.test → New → Java Class-> WebpageTest

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Message Shown by IntelliJ
"Do you want to add WebpageTest.java to Git: Click YES

WebpageTest.java

```
package org.test;

import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;
import org.testng.Assert;
import org.testng.annotations.AfterTest;
import org.testng.annotations.BeforeTest;
import org.testng.annotations.Test;

public class WebpageTest {

    WebDriver driver;

    @BeforeTest
    public void openBrowser() throws InterruptedException {
        driver = new ChromeDriver();
        driver.manage().window().maximize();
        Thread.sleep(2000);
        driver.get("https://binduprasad11.github.io/gradle-demo/");
    }

    @Test
    public void titleValidationTest() {
        String actualTitle = driver.getTitle();
```

BINDUSHREE V

```
        String expectedTitle = "Tripillar Solutions";
        Assert.assertEquals(actualTitle, expectedTitle);
    }

    @AfterTest
    public void closeBrowser() throws InterruptedException {
        Thread.sleep(1000);
        driver.quit();
    }
}
```

Replace URL with **your GitHub Pages URL**

Run Automation Test: gradle test. This will open the website

Task 4: Create JAR File

Update build.gradle
Replace content with:

```
plugins {
    id 'java'
    id 'application'
}

application {
    mainClass = 'com.example.Main'
}

jar {
    manifest {
        attributes 'Main-Class': 'com.example.Main'
    }
}
```

- Build in the intellij Idea and then go to bash, Make sure you are in root IdeaProjects/gradle-demo execute the following

```
./gradlew jar
java -jar build/libs/gradle-demo.jar
```

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```
BINDUSHREE V@Bindushree MINGW64 ~/IdeaProjects/gradle-demo (master)
$ ./gradlew jar
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::load has been called by net.rubygrapefruit.platform.i
nternal.NativeLibraryLoader in an unnamed module (file:/C:/Users/BINDUSHREE%20V/
.gradle/wrapper/dists/gradle-9.0.0-bin/d6wjpkvcgsg3oed0qlfss3wgl/gradle-9.0.0/li
b/native-platform-0.22-milestone-28.jar)
WARNING: Use --enable-native-access=ALL-UNNAMED to avoid a warning for callers i
n this module
WARNING: Restricted methods will be blocked in a future release unless native ac
cess is enabled


BUILD SUCCESSFUL in 2s
2 actionable tasks: 1 executed, 1 up-to-date
Consider enabling configuration cache to speed up this build: https://docs.gradl
e.org/9.0.0/userguide/configuration_cache_enabling.html

BINDUSHREE V@Bindushree MINGW64 ~/IdeaProjects/gradle-demo (master)
$ ls
build/        docs/    gradlew*     settings.gradle
build.gradle  gradle/  gradlew.bat  src/

BINDUSHREE V@Bindushree MINGW64 ~/IdeaProjects/gradle-demo (master)
$ cd build

BINDUSHREE V@Bindushree MINGW64 ~/IdeaProjects/gradle-demo/build (master)
$ ls
classes/  generated/  libs/  reports/  test-results/  tmp/

BINDUSHREE V@Bindushree MINGW64 ~/IdeaProjects/gradle-demo/build (master)
$ cd libs

BINDUSHREE V@Bindushree MINGW64 ~/IdeaProjects/gradle-demo/build/libs (master)
$ ls
gradle-demo.jar

BINDUSHREE V@Bindushree MINGW64 ~/IdeaProjects/gradle-demo/build/libs (master)
$ java -jar build/libs/gradle-demo.jar
Error: Unable to access jarfile build/libs/gradle-demo.jar

BINDUSHREE V@Bindushree MINGW64 ~/IdeaProjects/gradle-demo/build/libs (master)
$ cd ..

BINDUSHREE V@Bindushree MINGW64 ~/IdeaProjects/gradle-demo/build (master)
$ cd ..

BINDUSHREE V@Bindushree MINGW64 ~/IdeaProjects/gradle-demo (master)
$ java -jar build/libs/gradle-demo.jar
Hello Gradle!
```

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